Aviation Localization Implementation

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Abstract

Passive acoustic monitoring is an increasingly popular area of approach for the collection of bird population data. Birdwatching is a surprisingly popular hobby in the United States, enjoyed by roughly 1 in 3 American adults. By leveraging the popularity of this hobby, portable modules capable of passive acoustic monitoring can continue to be widely used, cost-efficient, and non-invasive to wildlife while improving the birdwatching experience. However, most citizen science approaches relying on automated monitoring are limited in dual-functionality when it comes to species identification and audio source localization. This project aims to design a low-power, passive, and portable sensing module that integrates machine learning with audio processing to automatically identify and approximate the location of nearby bird species. Bird vocalizations that typically range from 100 Hz to 10 kHz are captured using a custom tetrahedral microphone array designed to satisfy spatial Nyquist constraints for accurate localization within this bandwidth. Direction of arrival (DOA) estimation is performed using time difference of arrival (TDOA) measurements derived through the Generalized Cross Correlation with Phase Transform (GCC-PHAT) algorithm, providing robustness to environmental noise and amplitude variability common in outdoor settings. Species identification is achieved using BirdNET-Pi, a large language model that converts bird audio into spectrogram representations and applies a convolutional neural network to classify said birds. Localization and classification outputs are fused and transmitted to a mobile application for real-time visualization of detected species and their approximate azimuth and elevation. By providing both species identification and spatial estimation through this module, this system offers a practical tool for conservationists as well as hobbyist birdwatchers seeking enhanced situational awareness in the field.

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Introduction

The Bird Conservation Crisis and Citizen Science Opportunity

The decline of North American bird populations represents one of the most pressing conservation challenges of our time. Studies show that the United States has suffered a net loss of 3 billion birds in the past 50 years, with losses steadily continuing today [4]. Roughly 1 in every 3 U.S. adults engage in the hobby of birdwatching, which has supported 1.4 million jobs and generated $279 billion in total economic output in the United States [5]. This substantial community of engaged citizens represents a powerful resource for conservation data collection. Acoustic monitoring has emerged as of particular value, as it allows for the low-cost and non-invasive collection of data on the populations of birds. This will provide assistance to both the bird conservator and the hobbyist by lowering the level of difficulty in collecting data and encouraging beginner participation in conservation activities.

Project Overview and Approach

This project remedies the problem with the development of a low-power consumption, passive, and portable sensing module that combines the concepts of machine learning with audio processing in an automated manner to detect and estimate the presence of nearby bird species. The solution combines the concepts of:

1. **Custom Microphone Array and Spatial Localization**: A tetrahedral-shaped microphone array specifically created to record bird calls within the frequency spectrum of 100 Hz to 10 kHz while meeting the spatial Nyquist rate to precisely localize the sound within the same band. Direction of Arrival (DOA) calculation based on Time Difference of Arrival (TDOA) measurements obtained from the Generalized Cross Correlation (GCC) with Phase Transform algorithm to work efficiently in noise and intensity variation typical of outdoor conditions.
2. **Species Identification**: BirdNET-Pi, a large language model designed to transform the bird calls into spectrograms that its convolutional neural network (in particular, a ResNet chain) can analyze for species classification.
3. **Mobile Integration**: This is a React Native App that shows the localization and classification results. The app provides a live visualization of the detected species as well as the approximate AZ and EL angles.

A block diagram of the project can be shown in Figure 1 below. By providing species identification and spatial estimates, this technology provides conservationists surveying fields and birdwatchers wanting improved situational awareness. Indeed, because this technology operates on relatively low power and is portable, it can be used in the field for longer periods of time. Through this proof of concept and many other studies that incorporate localization and automation of species identification, passive acoustic monitoring will no longer be relegated solely to species detection. Instead, it will serve as the connection between the results of automation and active field observation.

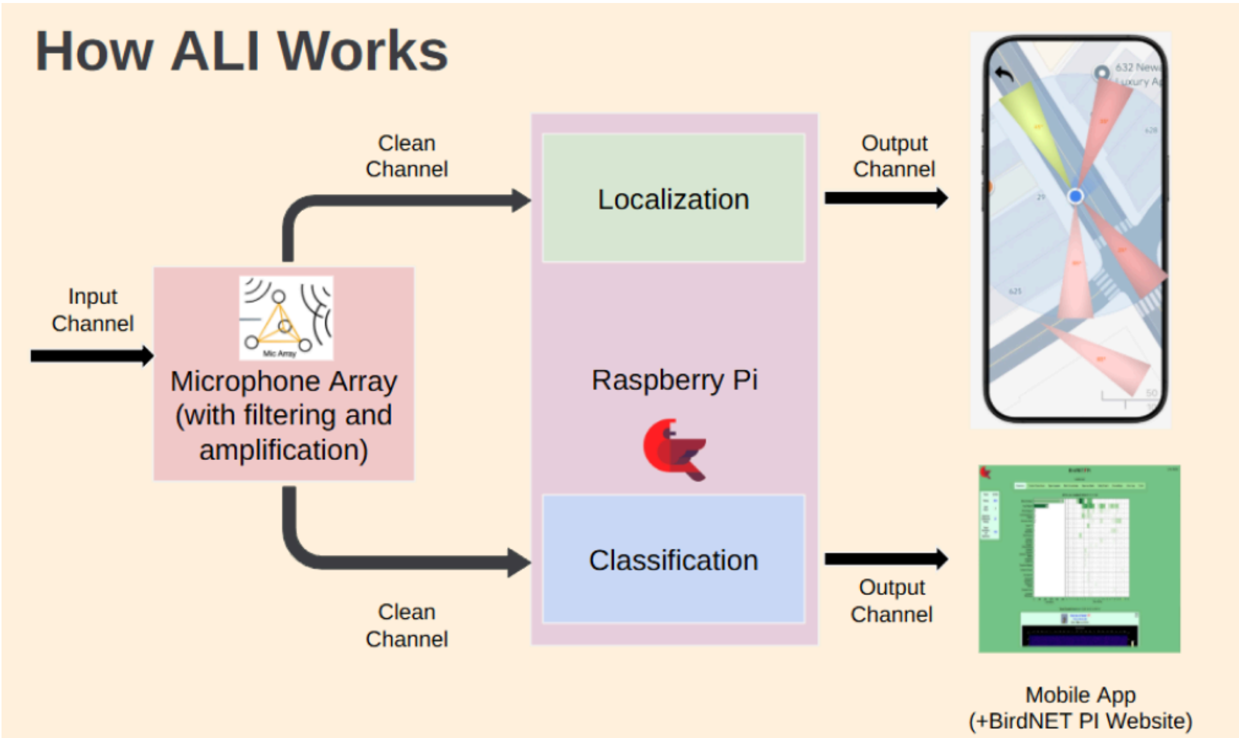

Figure 1. Block Diagram Showcasing Core Components of the Project

Backgrounds

Microphone Array

Plane Wave of Audio

A microphone array is a spatially distributed set of sensors that captures an incoming sound field in both time and space. Unlike a single microphone, which records only intensity, an array records the phase and amplitude differences of a propagating wave across multiple positions. After audio processing, these different measurements reveal the direction of arrival (DOA) of the sound source, enabling localization in two or three dimensions. Designing such an array requires defining the target frequency range, inter-microphone spacing, and geometry. All these parameters determine the array's spatial resolution, aliasing limits, and localizing capability. For starting audio processing, it is fair to assure that the sound source is sufficiently "far away", defined by

$$|r| > \frac{2L^2}{\lambda} \tag{1}$$

where r is the distance from the noise source and L is the length of the linear aperture. From here, a plane wave solution can be derived for the incoming audio wave:

$$x(t, r) = Ae^{j(wt-kr)} \tag{2}$$

where A is amplitude, $\omega=2\pi f$ is angular frequency, and k is the wave vector with magnitude $2\pi/\lambda$ and direction defined by azimuth ϕ and elevation θ. The relative phase difference at each microphone arises from $k * r_n$, where r_n is the position vector of the nth microphone. The steering vector

$$a(f,\ \theta,\ \phi)\ =\ [e^{-2\pi f\tau_1},\ e^{-2\pi f\tau_2},\ ...,\ e^{-2\pi f\tau_N}]^T \quad (3)$$

encapsulates the time delays $\tau_N = (r_n * k)/c$ for a given direction. This vector defines how the array's sensors "hear" a plane wave arriving from (θ,ϕ) and forms the mathematical foundation for beamforming and DOA estimation.

Spatial Sampling and the Aliasing Constraint

A key component of the microphone array design is to prevent aliasing, which is the introduction of unwanted or distorted audio signals from too low of sampling rate. To do this, the array has to follow Nyquist criterion,

$$f_s\ \geq\ 2\,f_{max} \quad (4)$$

where f_s is the sampling rate and f_{max} is the max frequency of your audio signal. Applying this to a microphone array picking up audio signals over air, which propagates at c = 343 m/s,

$$d \leq \frac{c}{2f_{max}} \quad (5)$$

where d is the distance between the microphones for the linear array. For a system operating over 100 Hz - 10 kHz, as typical of bird vocalizations, the spacing limit is approximately $d \leq 1.7$ cm. Arrays with wider spacing can achieve greater low-frequency directivity but will produce grating lobes, which are erroneous peaks in your signal, above the aliasing frequency $f_{alias} = c/(2d)$. In practical applications, a good compromise is 2 - 2.5 cm spacing for a good trade off between minimal high-frequency aliasing for improved 100–500 Hz response.

Far-Field and Near-Field Conditions

As stated earlier, most array theory assumes far-field propagation in which the audio source is sufficiently far from the aperture. However, when birds vocalize close to the sensors such as during flight or idle behavior, this assumption falls apart and the wavefront curvature produces amplitude differences across the array. Therefore, an analysis has to be done from the near-source perspective. The derivation for the weights of each microphone for a near-source audio stems similarly from that of the far-source signal, conclusion being:

$$w'_n(f) = \frac{d_n(r,\phi)}{d_0(r,\phi)} e^{j\frac{2\pi}{\lambda}[d_0 - d_n + nd\cos\phi]} \qquad (6)$$

with d_o being the distance from the source to the reference microphone and d_n being the distance from the source to the nth microphone. This adjustment is vital in ecological applications where birds may call from within a few meters.

Geometry and Dimensionality: 2D vs. 3D Arrays

The physical geometry of the array determines the dimensionality of localization. A linear or circular planar array estimates azimuth in two dimensions using time difference of arrival (TDOA) between microphones. However, such planar configurations suffer from elevation ambiguity , as multiple audio sources from differing elevations can yield identical TDOAs. To resolve both azimuth and elevation, the microphones must be arranged non-coplanar, such as tetrahedral, spherical, or multi-tiered “egg” geometries. 2D circular arrays (8–12 microphones spaced ≤ 2 cm) offer simpler 2D localization, while 3D arrays (12–16 microphones over multiple tiers) capture both elevation and azimuth at the cost of increased complexity.

Cardioid Elements and Differential Arrays

When array size is limited, low-frequency directivity can be improved by using cardioid microphones with first-order differential sensors combining pressure and pressure gradient responses. The general cardioid pattern for microphones follows the equation,

$$P(\theta) = 0.5(1 + \cos\theta) \tag{7}$$

which attenuates rear sound by 6 dB. This reduced background noise can be used to combine multiple cardioid elements in a differential or circular configuration to boost directivity without violating the spacing constraint for 10 kHz operation. In bird-localization arrays, this approach maintains sensitivity to 100–500 Hz calls and wingbeat sounds while preventing spatial aliasing at higher harmonics.

Audio Processing

Direction of Arrival Estimation and Array Signal Processing

Direction-of-arrival (DOA) estimation is the process by which a microphone array determines the spatial origin of a sound source. Unlike a single microphone that captures only the temporal characteristics of sound (amplitude and frequency over time), a microphone array exploits spatial diversity. When a sound wave propagates from a distant source, it arrives at different microphones at slightly different times and with varying phase relationships [1]. The Generalized Cross-Correlation (GCC) method is a widely used technique in signal processing for estimating the time difference of arrival (TDOA) between multiple sensors receiving the same signal. In using this method, the times a sound is received by each microphone is determined. Differences between these times are calculated and then used to find the direction of the source of the sound. For simplicity, we will examine the algorithm using just two sensors. Cross correlation is defined as follows:

$$R_{12}(\tau) = \int_{-\infty}^{\infty} x_1(t)\, x_2(t+\tau)\, dt. \tag{8}$$

This is a measurement of how similar the two signals are when one is shifted in time by the amount τ, and the estimated delay (TODA) can be computed as

$$D = \arg\max R_{12}(\tau). \tag{9}$$

Like most signal processing methods, it is beneficial to examine signals in the frequency domain. Fourier transforms are applied to received signals and then their cross-power spectrum (the Fourier transform pair of the cross correlation) is calculated.

$$G_{12}(f) = X_1(f) X_2^*(f) \tag{10}$$

Next, in order to filter out unwanted noise from our analysis, a filter is applied in the form of a weighting function Ψ(f). The filter is applied to the cross-power spectrum and the result is converted back into the time domain via the inverse Fourier transform..

We are specifically interested in the Phase Transform (PHAT) method, so we will use the following weighting function.

$$\Psi(f) = \frac{1}{|G_{12}(f)|} \tag{11}$$

This leaves just the phase information. Because we plan on implementing the algorithm outdoors where acoustic conditions may vary, we cannot rely on amplitude information as much as the phase information. Thus, focusing on the phase information will be beneficial. Filtering the cross-power spectrum, we get the following result.

$$G_{12}^{\mathrm{PHAT}}(f) = \frac{X_1(f)X_2^*(f)}{|X_1(f)X_2^*(f)|} \tag{12}$$

Converting back to the time domain yields the following expression:

$$R_{12}^{\mathrm{PHAT}}(\tau) = \int_{-\infty}^{\infty} G_{12}^{\mathrm{PHAT}}(f)\, e^{j2\pi f\tau}\, df[1]. \tag{13}$$

In order to go from TDOA to DOA, a few computations are performed. These computations are outlined below, where c denotes the speed of wave propagation: in this case, the speed of sound.

The two microphones are separated by a distance d, and the angle θ is the angle at which the source of the signal is located (DOA).

$$D = \frac{r \cos\theta}{c} \tag{14}$$

Because this implementation requires converting an analog signal (sound) into a digital one, in order to do computations, we must take into account the sampling rate R.

$$D = \frac{d\cos\theta}{c} R \tag{15}$$

$$\theta = \arccos\left(\frac{cD}{dR}\right) [2] \tag{16}$$

When using p microphones,

$$R_{12}(\tau) = \sum_{m=1}^{p} \sum_{n=1}^{p} \int_{-\infty}^{\infty} G_{12}^{\mathrm{PHAT}}(f)\, e^{j2\pi f(\tau_m - \tau_n)}\, df[1]. \tag{17}$$

In summary, GCC is implemented in the frequency domain. The microphone signals are transformed using the Fourier transform, and their cross-power spectrum (how correlated the signals are at each frequency) is computed. To make the method work well with real-world, noisy, and broadband sounds (like bird calls), a weighting function is applied before transforming back to the time domain. One effective weighting is the Phase Transform (PHAT), which normalizes each frequency component by its magnitude. The resulting GCC–PHAT function typically shows a clear peak at the time delay corresponding to the true direction of sound.

Spatial Sampling and the Aliasing Constraint

Just as temporal sampling of audio signals must satisfy the Nyquist criterion to avoid aliasing ($f_s \geq 2f_{max}$), spatial sampling in microphone arrays must obey an analogous constraint. For a linear array with element spacing d, the maximum unambiguous frequency is limited by equation (5) where $c \approx 343$ m/s is the speed of sound in air. Arrays with wider spacing can achieve greater low-frequency directivity but will produce grating lobes, erroneous peaks in the spatial spectrum, above the aliasing frequency. [1]

Deep Learning for Acoustic Species Identification

Modern bird identification systems leverage deep learning to automatically classify species from acoustic recordings [8]. The process begins by converting audio signals into spectrograms—two-dimensional representations where one axis represents time, one axis represents frequency, and brightness indicates amplitude [9]. This is done in order to convert received audio signals into images, which neural network architectures have been shown to process more reliably [8].

In the field of deep learning, Convolutional Neural Networks (CNNs) are the typical choice for image processing. The main block of a CNN is the convolutional layer that outputs a number of cells, with each cell being a function of a local neighborhood of input cells. The layer consists of a matrix multiplication, an offset by a vector, and an activation function (typically a ReLU that zeros negative values). The matrices and offset vectors used in convolutional layers are determined through gradient descent, tuning them so that the network produces correct outputs for labeled training data [10].

Advanced systems employ Residual Networks (ResNets), which enable hundreds of convolutional layers to work in series. ResNets use residual layers where the input is directly added to the output of convolutional operations through skip connections. This design keeps the derivative close to 1, drastically simplifying chain rule calculations involved in gradient descent for larger models [6]. This allows ResNet stacks to be much larger and more sophisticated without a significant increase in computational expense.

The network outputs a probability vector where each entry represents the likelihood that the input corresponds to a particular species. Advanced systems incorporate metadata such as geographic location and season to apply occurrence masking, setting probabilities to zero for species unlikely to be present at that location and time. This metadata-informed approach improves classification accuracy by incorporating ecological knowledge into the pipeline [8].

Mobile App Integration

The development of the sound localization interface aims to overlay directional vectors on a geographic map, displaying:

- **Azimuth and elevation cones**: Visual representation of the estimated bird location as directional wedges or rays emanating from the user's position
- **Geographic context**: Real-time map integration showing terrain and location provided by React Native Map feature
- **Multiple simultaneous detections**: Ability to display and differentiate between multiple calling birds with distinct directional indicators
- **Historical tracking**: Optional trail or persistence view showing the sequence of detections over time

Figure 2. Proposed Localization Visualization Interface

Related Work

In 2023, a group from Japan used a single 16-channel microphone array to estimate the azimuth and elevation angles of bird calls in order to characterize how birds use their acoustic environment. The 16-channel microphone array is arranged in an egg shape and was named DACHO (meaning ostrich in Japanese). The module has captured forest bird vocalizations and estimated the DOA of sounds in both azimuth (horizontal angle) and elevation (vertical angle). For localization, they processed the multi-channel recordings using the HARK (Honda Research Institute Japan Audition for Robots with Kyoto University) software framework and its HARKBird extension, which implements SEVD-MUSIC (Standard EigenValue Decomposition Multiple Signal Classification) to compute DOA from the array data and GHDSS (Geometric High-order Decorrelation-based Source Separation) to extract separated sound events for further analysis. The group evaluated the localization method by replaying bird calls from a loudspeaker elevated above the array at various horizontal distances in forest conditions, finding that the system could estimate azimuth with errors of 5° or less and elevation with errors of 15° or less in most cases when the source was within about 35 m of the array. [14]

This research is very relevant to our project, since this group is using a compact microphone array to localize bird calls in three dimensions. Although DACHO is able to localize both the azimuth angle and elevation, it is unable to identify each individual bird automatically. Of course, consulting an expert is great for research purposes, our intended audience leans more toward novices in bird identification who do not have fast access to an expert. Still, DACHO succeeds in a large portion of what we aim for. Our group will be looking into the HARKBird algorithm to perhaps adapt it for our use.

Earlier this month, a pair from the University of Hohenheim published a paper describing a system which both identifies and localizes birds based on their calls. The pair's approach combines multiple cost-effective autonomous recording units with a machine learning model (BirdNET) to both identify species and determine the spatial origin of their vocalizations. Six recording devices were placed between 25 to 100 meters apart from each other to record bird calls. For each possible time shift, they calculated a similarity measure defined as the sum of the product of spectrogram amplitudes across time and frequency bins, allowing frequency bands to be selectively filtered beforehand. To reduce noise, similarity was smoothed using a moving average across three neighboring time shifts, and the time shift with the highest similarity was taken as the estimated delay between recordings. A quality metric called "similarity gain," defined as the relative difference between the maximum similarity and the mean similarity across all tested shifts, was computed to assess confidence in each delay estimate. In validation tests with replayed bird sounds in an agricultural landscape, after filtering out low-confidence identifications, the algorithm localized over 92 % of sounds within 2 m of their true positions (with about 78 % within 1 m), demonstrating high accuracy. [15]

This system is similar to ours in the sense that it is both localizing and identifying birds based on their calls using BirdNET. The major difference between our goal and this system is that this system is not compact. Because we are targeting individuals who are beginners in bird identification, we want the module to be portable and easy for casual use. The method of scattering microphones is not ideal for our use case. However, this paper informs us of methods to process the signals that we collect.

Project Description

By break-up, from physical parts, our device consists of a microphone array for capturing bird vocalizations, a Raspberry Pi 4B to make the identification and classification, and ADCs. Thus, the microphone array consists of four cardioid microphones in a tetrahedral arrangement. Signals from these microphones are then routed to the ADCs where they get converted into digital format and are ready to be fed to the Raspberry Pi. Running on the Raspberry Pi are two algorithms-a free, open-source machine learning model called BirdNET for identification and our original implementation of GCC-PHAT for localization.

Once our device has calculated the direction and species of a bird, this is then transmitted to a mobile device and displayed using a mobile application. This mobile application was developed in order to provide field-accessible interaction with the Bird Identification and Localization system. A critical design challenge arose from conflicting network requirements: the BirdNET Pi system operates as a local web server accessible via WiFi, but field deployment often requires mobile devices to be able to function as cellular hotspots for internet connectivity. A mobile device cannot simultaneously operate as a cellular hotspot and connect to the WiFi network provided by the Raspberry Pi to access the local web interface when serving as a hotspot. In order to overcome this fundamental networking limitation, a dual-mode architecture has been implemented. In the internet-connected mode, Tailscale VPN forms a secure overlay network that maintains persistent communications between the mobile device and the Raspberry Pi regardless of underlying network topology, thereby allowing simultaneous cellular hotspot operations and access to BirdNET Pi. In offline mode, the mobile device directly connects to the

WiFi access point created by the Raspberry Pi, forgoing internet connectivity in favor of enabling totally autonomous field operations without cellular coverage.

Implementation of the core navigation architecture of the mobile application was successful; users can access the main two system functions seamlessly. A select menu in the main interface navigates users either to the sound localization view or to the BirdNET Pi web interface, as shown in Figure 3. Integration testing proved successful that the mobile app could reliably connect and display the BirdNET Pi local website, successfully validated to operate in internet-connected mode using Tailscale VPN, and offline mode using a direct WiFi connection to the Raspberry Pi.

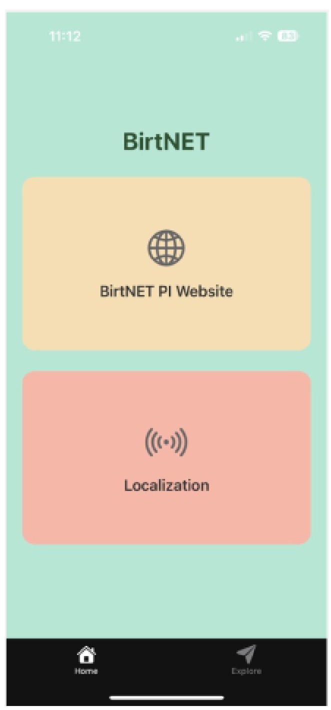

Figure 3. Mobile Application Main Navigation Interface

Ethical Implications

Although the technologies used and developed in this project may seem harmless at first, there are various ethical issues that are worth considering as we continue to develop ALI. One such consideration is the unintentional recording of human conversations. Although the purpose of this project is to record and process bird calls, these bird calls can only be detected by recording all surrounding audio and identifying which parts of this recorded audio come from bird calls using BirdNET-Pi. As such, the audio that will be recorded by our device can come from any source, including nearby people engaging in conversation. This would be unlikely in the wildlife setting that our device is intended to be used in and our device would do nothing with these hypothetical recorded human conversations. However, the fact remains that unless measures are taken, scenarios are possible where people's conversations are recorded by our device without their consent. This may be considered a breach of privacy, therefore it is important for our device to discard any and all information that is not used for its intended purposes.

Even then, our device may still be used in ways that can be considered harmful. For example, with the information of the direction and species of a nearby bird, poachers may be able to track down and hunt endangered bird species, unintentionally worsening the issue of wildlife endangerment. This issue is much more difficult to provide a countermeasure for, since it is a direct cause of the delivery of the information our device is designed to deliver. One potential solution may be to leverage the metadata processing in BirdNET to prevent our device from identifying endangered bird species, but collection of this data can also be important and this solution fails to protect non-endangered species.

Status

Below is the Gantt chart for our project to be completed. The green checks denote completed tasks.

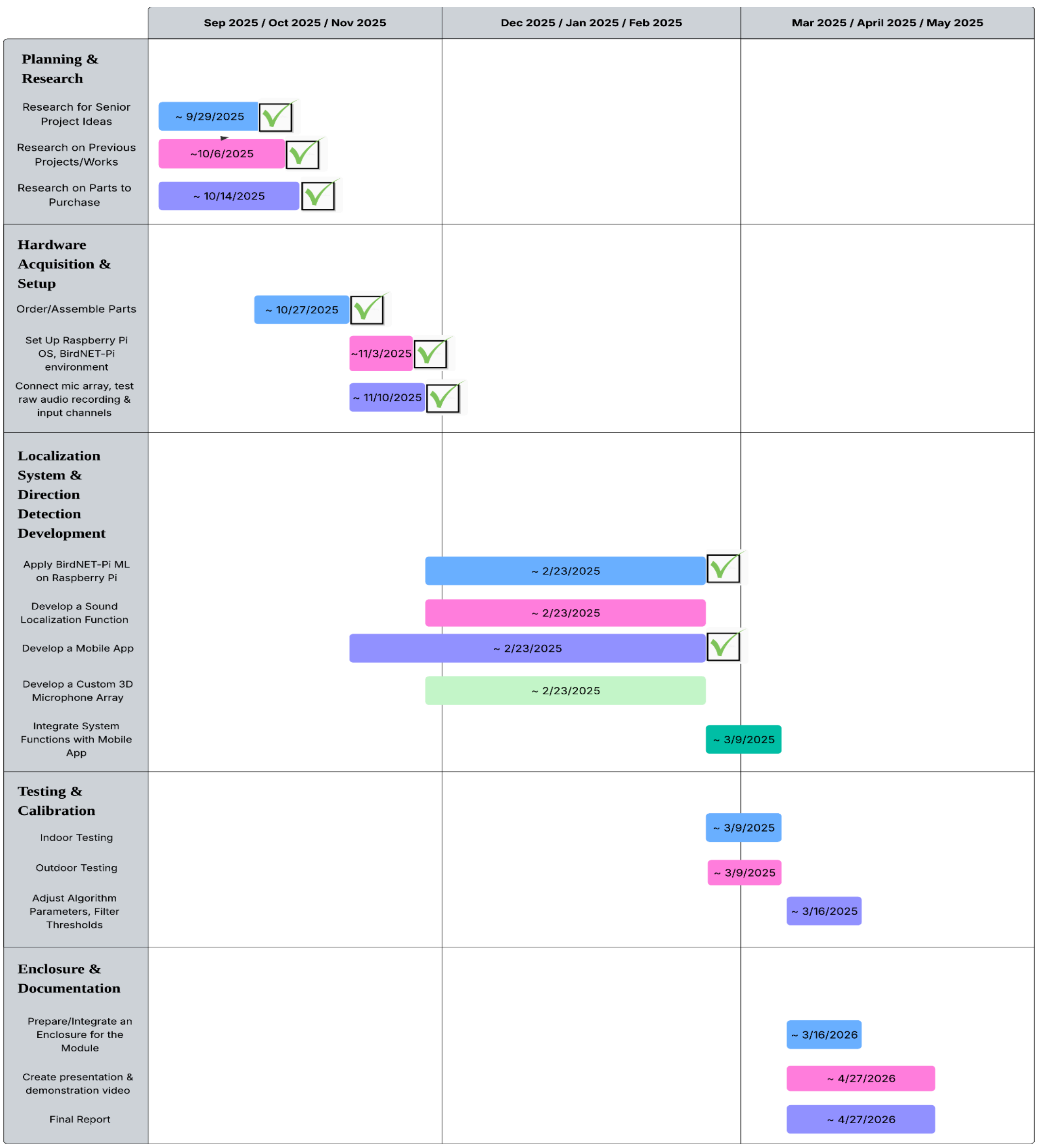

Figure 4. The Gantt chart

Alvee

Researched the background for optimal microphone design for audio localization including multiple times of microphones such as omnidirectional, bidirectional, and cardioid microphones in tandem with microphone array structures. Research led to the conclusion that cardioid microphones were best used in tandem with a tetrahedral structure for 3D localization as at least 4 microphones are needed to calculate both azimuth and elevation angles for an audio source. Furthermore, researched tetrahedral microphone design based around Nyquist Sampling Theorem, leading to design parameters regarding spacing for our custom built microphone array. Now that the cardioid microphones have been purchased, future steps will include physically assembling the array and designing a case for the structure.

Anna

Setup the BirdNET-Pi environment on the Raspberry Pi, and tested that it worked with a USB microphone. The testing was done by playing prerecorded bird calls on a speaker near the microphone. Researched source localization algorithms which can work for multiple sources. Settled on GCC-PHAT. Future work will include writing up the algorithm to conduct the source localization. Additionally, Anna will be configuring the Raspberry Pi to work with the microphones using the analog to digital converters (ADC) that we have purchased which communicate using the SPI protocol.

Darius

Performed background research for existing work and project motivations. Learned how BirdNET works and may look into ways of improving it. Will be making CAD files for the microphone array as well as the enclosure for the final product. Will also be working on ensuring the signal from the correct microphone is delivered to BirdNET for classification and assisting with the implementation of the localization algorithm.

Burt

Development of the sound localization interface for the mobile app is currently on hold pending completion of the hardware and algorithmic components. Since the localization algorithm implementation is still in progress alongside the construction of the 3D tetrahedral microphone array, we have strategically paused mobile app development to avoid premature interface design that may not align with the final data format and performance characteristics of the DOA estimation system. During this period, Burt has contributed to the development of the custom microphone array by researching I²C and SPI protocol implementation between the array and the Raspberry Pi.

Once the microphone array hardware and GCC-PHAT localization algorithm are operational, Burt will resume development by implementing an integrated mapping interface.

Conclusion

Aiding bird conservation requires a level of expertise in bird identification. There are over 700 bird species in North America alone [16], experts in one region can easily become novices in another. Many times, there are specific birds of interest and one way to find them is through listening for their calls. However, it is very difficult to identify a specific call from among many others without expertise. Acoustic monitoring has become an especially useful tool because it allows for non-invasive, cost-effective data collection, lowering barriers for both professional conservationists and amateur birdwatchers. We propose a low-power, and portable acoustic monitoring system that combines machine learning with audio signal processing to detect and localize bird species automatically. A tetrahedral microphone array captures bird calls in the 100 Hz to 10 kHz range while satisfying the spatial Nyquist criterion for precise localization. Time difference of arrival (TDOA) measurements are computed using the generalized cross-correlation with phase transform (GCC-PHAT) algorithm, which performs reliably under outdoor noise and variable sound intensity. Species classification is handled by BirdNET-Pi, which transforms recorded calls into spectrograms analyzed by a convolutional neural network based on a ResNet architecture. Finally, a React Native mobile app which visualizes the results in real time, showing detected species along with estimated azimuth and elevation angles.

By providing both species identification and spatial information, this system enhances situational awareness for birdwatchers and supports detailed field surveys for conservationists. The portable, low-power design allows for both casual and extended deployment in natural habitats. More broadly, this project demonstrates how passive acoustic monitoring can move beyond basic species detection. It links automated analysis with active field observation, empowering citizen scientists and professionals alike to contribute meaningfully to bird conservation.

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